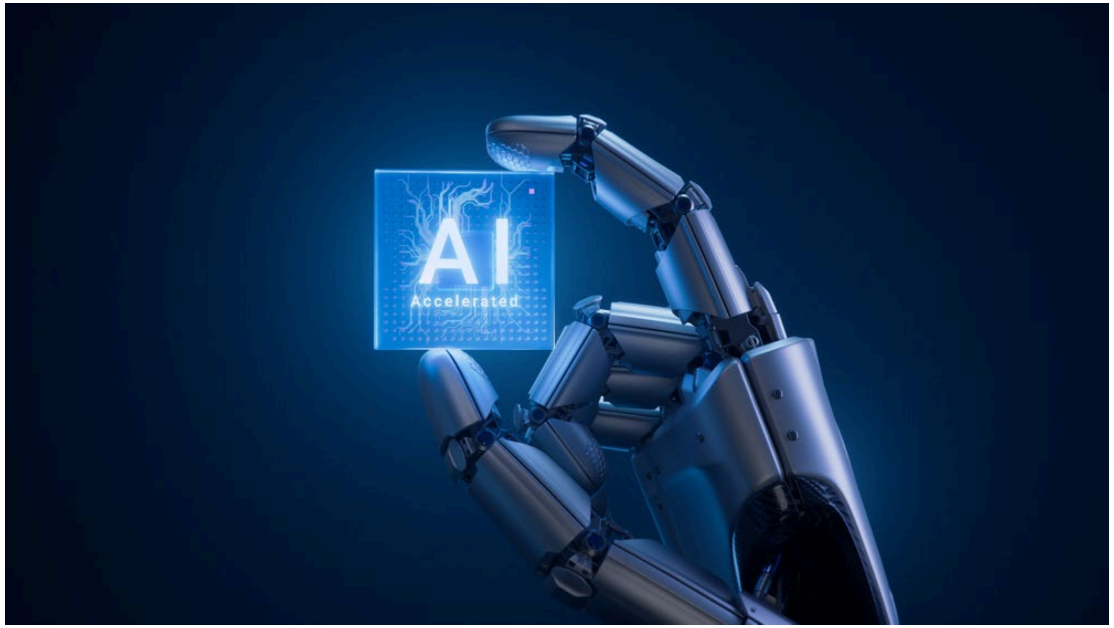
CS 600: Artificial Intelligence Foundations

Foundations of Artificial Intelligence

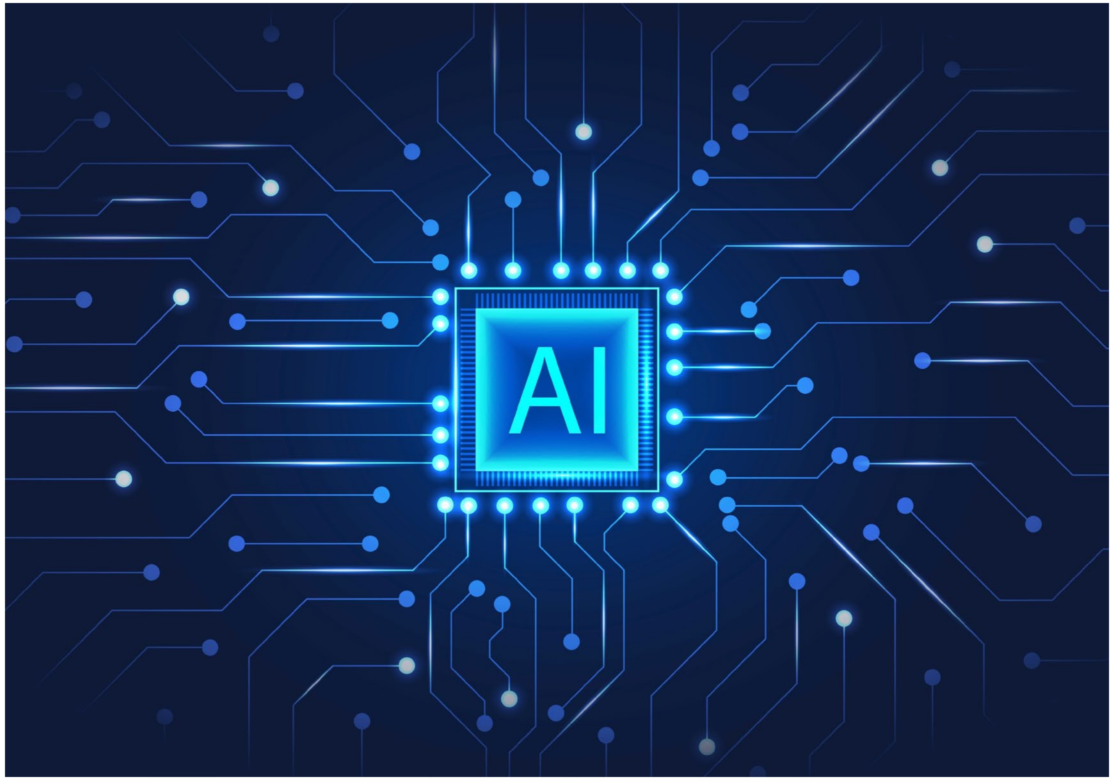
What Is Artificial Intelligence?

- Artificial Intelligence (AI) is a branch of computer science
- Focuses on creating systems that can perform tasks that normally require human intelligence
- Examples:
    - Voice assistants (Siri, Alexa)
    - Recommendation systems (Netflix, Amazon)
    - Spam filters and navigation apps



Why AI Matters?

- Helps analyze large amounts of data
- Automates tasks
- Supports decision-making
- Used in everyday technology, often without us noticing it



Core Ideas Behind AI

- AI systems are built around a few key concepts:
  - Intelligent agents
  - Data and knowledge
  - Learning and adaptation
  - Reasoning and decision-making

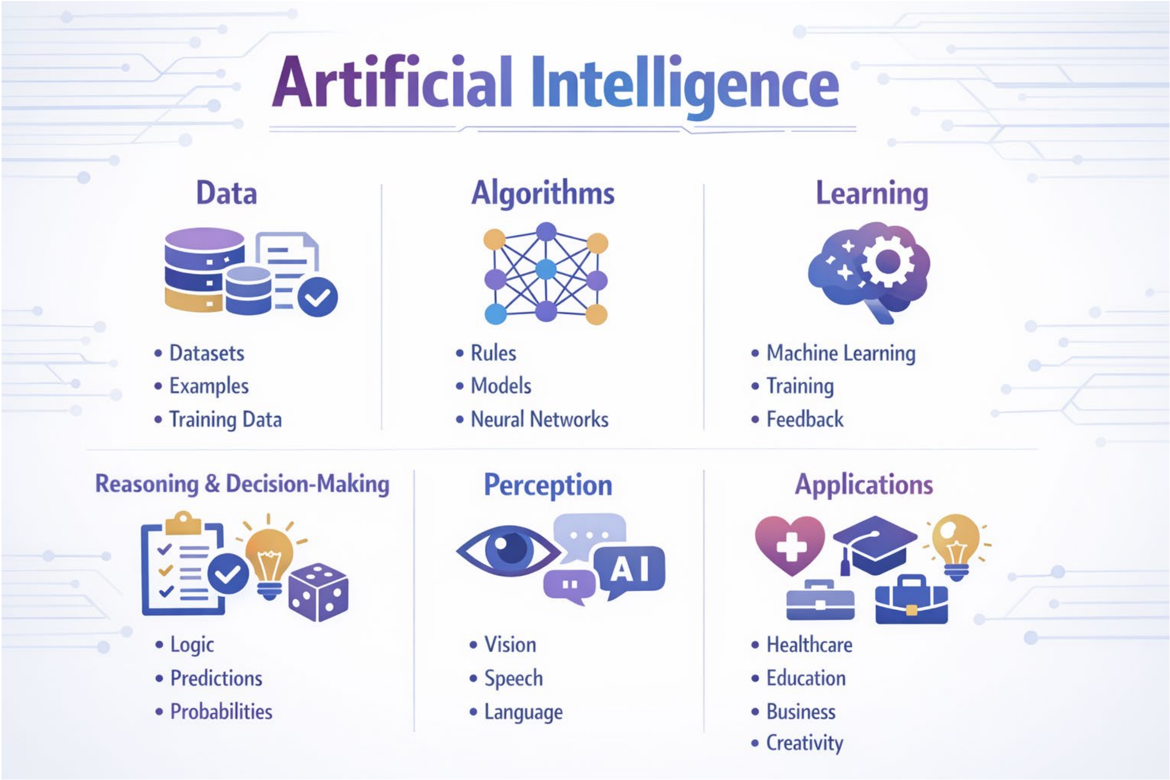


Image created using ChatGPT

Intelligent Agents

- An intelligent agent:
  - Observes its environment
  - Makes decisions
  - Takes actions to achieve a goal
- Examples:
  - Self-driving car
  - Chess-playing program
  - Recommendation system

Data and Knowledge

- AI systems rely on:
  - Data (examples, information, experiences)
  - Knowledge (rules, relationships, patterns)
- Two common approaches:
  - Rule-based systems (written by humans)
  - Data-driven systems (machine learning)

Learning and Adaptation

- Many AI systems can learn over time
- Learning means:
  - Improving performance with experience
  - Finding patterns in data
- Example:
  - Streaming services are learning your preferences

Reasoning and Decision-Making

- AI systems must choose the best action
- This may involve:
  - Logic and rules
  - Comparing options
  - Using probabilities
- Goal: Make the best possible decision with available information

Types of Artificial Intelligence

- Two major categories:
  - Narrow (Weak) AI
  - General (Strong) AI

Narrow (Weak) AI

- Designed to perform a specific task
- Examples:
  - Voice assistants
  - Facial recognition
  - Spam filters
  - Recommendation systems
- All AI in use today is Narrow AI

General (Strong) AI

- Would be able to perform any intellectual task a human can
- Could learn and reason across many areas
- Does not currently exist
- Remains a long-term research goal

Why These Foundations Matter

- Helps us:
  - Understand how AI really works
  - Recognize its limits
  - Use AI responsibly
- AI is not magic:
  - Built by humans
  - Trained on data
  - Guided by algorithms

Quiz

Click the **Quiz** button to edit this object

What is an intelligent agent in Artificial Intelligence? Select the correct answer option:

- A database that stores large amounts of information
- A computer program that only follows fixed rules
- A machine that looks like a human
- A system that observes its environment, makes decisions, and takes actions to achieve a goal

Summary

- AI systems can:
  - Observe
  - Learn
  - Reason
  - Act
- Foundations of AI prepare us for:
  - Machine learning
  - Neural networks
  - Ethical considerations in AI

Closing

- **Next up:**
  We’ll build on these foundations by exploring how machines learn and how data drives modern AI systems.